04 / Grounding

What connects the representation or interpretation to physical evidence?

EX02 / UMI | Source analysis | C1 | P: paper; W: project; R: repository

Prediction / representation	Matched reference	Difference / ambiguity	Conclusion / scope limit

1. Comparison boundary

Name the prediction/measurement and reference. Match conditions; identify simulated versus physical evidence.

Match demonstration and deployment observation/action semantics, especially pose coordinates and timing. [P Policy design]

2. Quantities and validity

Declare quantity, units, frame, tolerance and measurement limits. Distinguish a chosen tolerance from observed uncertainty.

Latency is a time offset; relative poses need declared frames. These are distinct calibration problems. [P App A]

3. Results and uncertainty

Give paired outcomes, trial counts and uncertainty or limits. If untested, record the protocol and missing evidence explicitly.

Observation streams are aligned to camera timestamps; measured execution delays determine when future actions are sent. [P Policy design]

4. Interpretation and challenge

Bound the conclusion and consider another cause of the difference. Choose a diagnostic test or scope restriction.

SLAM error or lighting shift can break transfer; correct timing alone does not guarantee task success. [R limitations]

Review: Could sensing error, reset differences or selection explain the discrepancy instead of model error?

Example response: Calibration, SLAM error and lighting can each cause a transfer discrepancy.

Worksheet 4 / Grounding - evidence and team review

EX02 / UMI | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX02 Project: UMI Claim ID: C1 Revision: draft 1

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: paper sections and public documentation; no dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/abs/2402.10329v1	Paper: method/results; reported evidence, not our measurement.
W	real.stanford.edu/umi	Project: system/videos; selected examples do not establish reliability.
R	github.com/real-stanford/universal_manipulation_interface	Repository: documented route; code and data not executed or audited.

Next action / owner / verification:

Calibration, SLAM error and lighting can each cause a transfer discrepancy. Reviewer: inspect the cited basis before adapting this answer.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

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8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.